

Manifold-Informed Architecture Benchmark

CIFAR10: 10 classes, 3,072D input

Generated: 2026-08-04 03:29:24 | Apple M5 Max MacBook Pro, 64 GB RAM, 2TB SSD

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Best: Standard (1024→512) test acc = 0.5167 ± 0.0075

Manifold: 3,072D → 34D (98.9% noise dimensions)

Experimental Setup

Parameter	Value
Dataset	CIFAR10
Input dimensionality	3,072D
Classes	10
Intrinsic dim (d)	34
Variance threshold (τ)	0.9
Epochs	60
Trials	5
Batch size	512
Learning rate	0.001
k neighbors (local PCA)	50
Discovery samples	500
Samples per class	10

Manifold Discovery

Local PCA over the training set, $k = 50$ neighbors. Intrinsic dimensionality d is taken as the maximum per-class maximum at $\tau = 0.9$, clamped to $n_{\text{classes}} = 10$.

τ	Mean d	Std	Min	Max	Noise (%)
0.95	36.0	1.8	24	40	98.8
0.90	28.8	1.9	18	33	99.1
0.85	23.6	1.9	14	28	99.2
0.80	19.6	1.8	11	24	99.4

Per-Class Intrinsic Dimensionality

Class	Mean d	Std	Min	Max
frog	31.6	1.8	28	33
truck	31.4	1.6	28	34
horse	31.1	0.8	30	33
automobile	30.8	2.2	25	33
deer	29.2	1.5	28	32
dog	28.4	0.8	27	30
cat	27.8	1.0	26	29
bird	27.7	2.1	25	31
airplane	26.5	1.2	25	29
ship	25.6	1.6	22	27

Architecture Comparison

Architecture	Params	Test Acc ($\mu \pm \sigma$)	Loss	Acc/Kparam
Standard (1024→512) ★	3,676,682	0.5167 ± 0.0075	4.4642	0.0001
Wide Manifold (d+1, d=34)	107,915	0.4558 ± 0.0038	1.6858	0.0042
Manifold (d=34)	104,832	0.4585 ± 0.0026	1.6821	0.0044
Manifold + ManifoldAdam (d=34)	104,832	0.4731 ± 0.0035	1.4881	0.0045
ManifoldAdam (1024→512, proj→34D)	3,676,682	0.4623 ± 0.0094	3.3072	0.0001
PCA→34D + MLP (2d→d)	5,076	0.4912 ± 0.0046	1.4503	0.0968
Intrinsic Dim (PCA→34D→output)	1,540	0.4675 ± 0.0051	1.4954	0.3036

★ = best performing architecture

Key Findings

- **Best architecture:** Standard (1024→512) — test accuracy 0.5167 ± 0.0075
- **Manifold compression:** 3,072D → 34D (98.9% of ambient dimensions carry no structure)

Result Figure

Provenance

Item	Value
Repository	Flux-Frontiers/waverider
Commit	3c5d283 (-abbrev-re 3c5d28344fd5c5ace0b3399593d61b40363afcd)
Commit date	2026-08-04 03:22:44 +0000
Commit message	docs(readme): add holographic-rendering Breaking News section; audit fixes
Python	3.12.3
TensorFlow	2.21.0
Device	CPU (forced)
Host	vm
OS	Linux-6.18.5-fc-v18-x86_64-with-glibc2.39
Machine	Apple M5 Max MacBook Pro, 64 GB RAM, 2TB SSD
Generated	2026-08-04 03:29:24

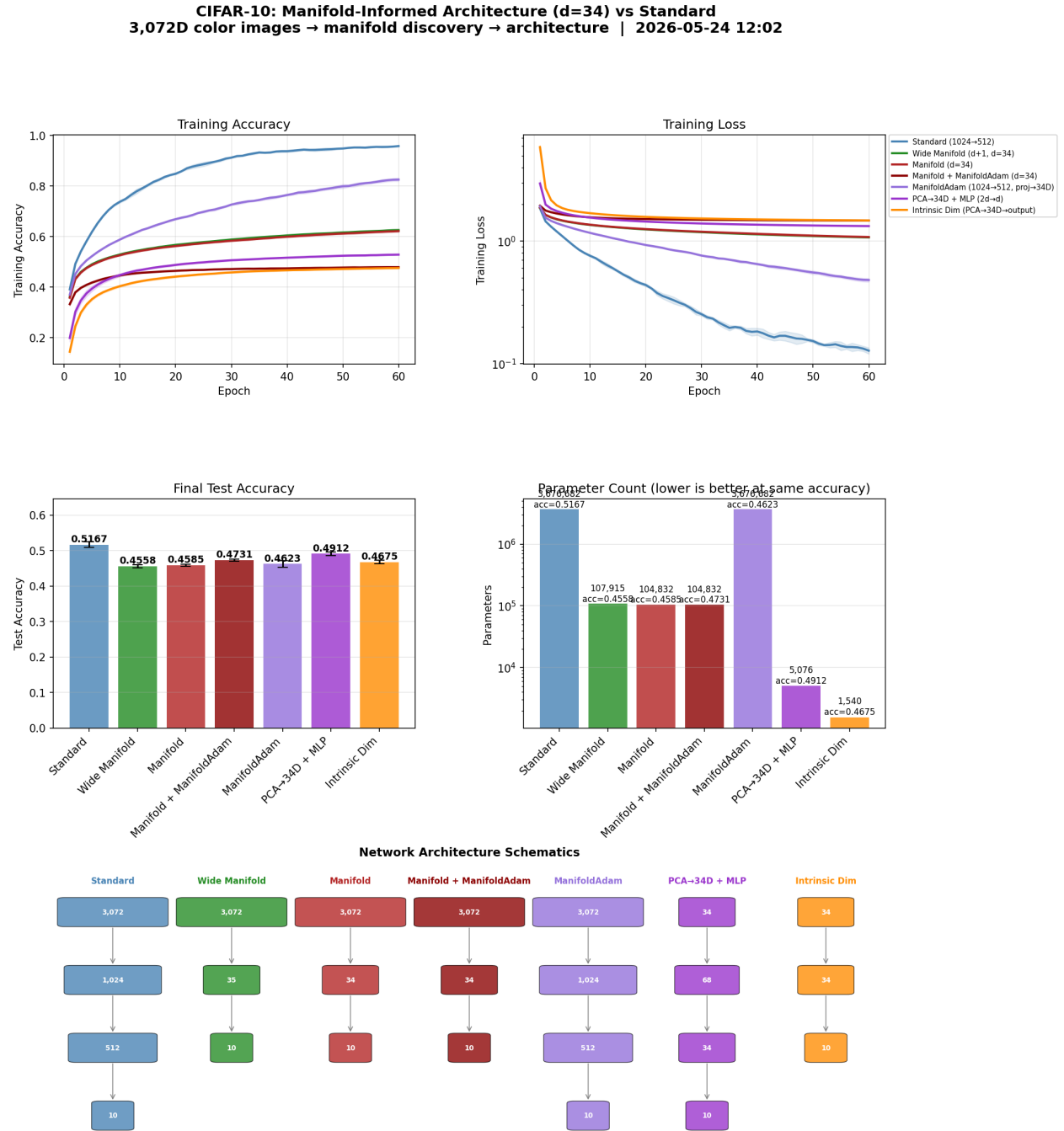


Figure 1: CIFAR10 manifold-informed vs. standard architecture comparison. $d = 34$, 60 epochs, 5 trials.